

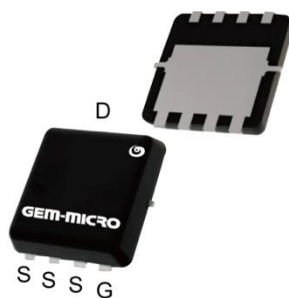
65V/75A N-Channel Advanced Power MOSFET

PRODUCT SUMMARY

V_{DS}	I_D	$R_{DS(on)}$ m Ω Max
65V	75A	6 @ $I_D=40A$, $V_{GS}=10V$
		10 @ $I_D=20A$, $V_{GS}=4.5V$

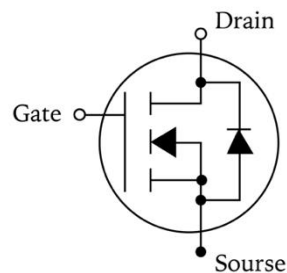
Features

- Enhancement mode
- Low on-resistance $R_{DS(on)}$ @ $V_{GS}=4.5V$
- New SGT-MOS Technology
- 100% Avalanche test
- Pb-free lead plating; RoHS compliant



Pin Assignment & Symbol
8-Lead Plastic: PDFN 5*6

Pin1-3: Source
Pin4: Gate
Pin5-8: Drain



Absolute Maximum Ratings ($T_A=25^\circ C$, unless otherwise noted)

Symbol	Parameter		Rating	Unit
V _{DS}	Drain-source voltage		65	V
V _{GS}	Gate-source voltage		±20	V
I _S	Diode continuous forward current	T _C =25°C	75	A
I _D	Continuous drain current @V _{GS} =10V	T _C =25°C	75	A
I _D	Continuous drain current @V _{GS} =10V	T _C =100°C	48	A
I _{DM}	Pulse drain current tested ①	T _C =25°C	300	A
I _{DSM}	Continuous drain current @V _{GS} =10V		23	A
I _{DSM}	Continuous drain current @V _{GS} =10V	T _A =70°C	18	A
E _{AS}	Maximum avalanche energy, single pulsed ③		25	mJ
P _D	Maximum power dissipation	T _C =25°C	45	W
P _{DSM}	Maximum power dissipation ②		4.2	W
T _j , T _{stg}	Operating junction and storage temperature range		-55 to 150	°C

Thermal Characteristics

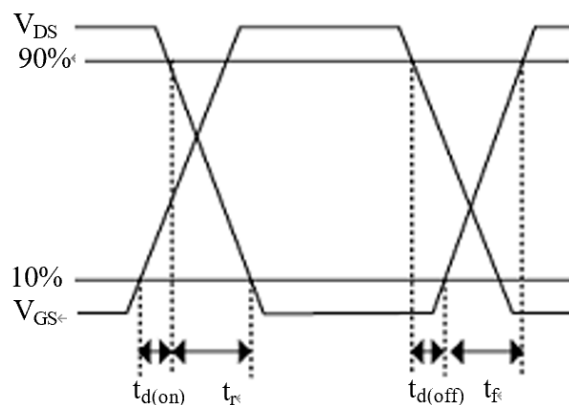
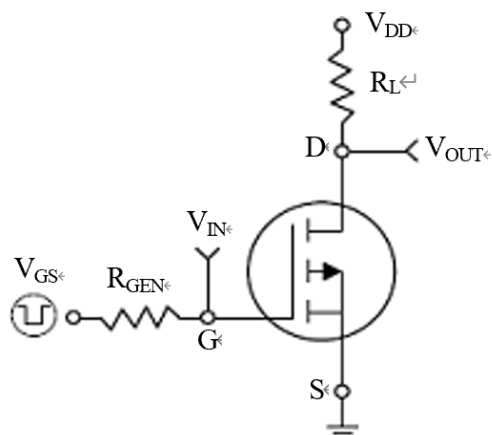
Symbol	Parameter	Typ	Max	Unit
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Electrical Characteristics (T_A=25°C, unless otherwise noted)

Symbol	Parameter	Condition	Min.	Typ.	Max.	Unit
Static Electrical Characteristics (unless otherwise stated) @ T_J=25°C						
BV _{DSS}	Drain-source breakdown voltage	I _D =250μA, V _{GS} =0V	65			V
I _{DSS}	Zero gate voltage drain current	V _{DS} =60V, V _{GS} =0V			1	μA
I _{DSS}	Zero gate voltage drain current	V _{DS} =60V, V _{GS} =0V			100	μA
I _{DSS}	Zero gate voltage drain current	V _{DS} =60V, V _{GS} =0V			100	μA
I _{GSS}	Gate-body leakage current	V _{DS} =0V, V _{GS} =±20V			±100	nA
V _{GS(th)}	Gate threshold voltage	I _D =250μA, V _{DS} =V _{GS}	1.3	1.8	2.5	V
R _{DS(on)}	Drain-source on-state resistance	I _D =40A, V _{GS} =10V		4.5	6	mΩ
R _{DS(on)}	Drain-source on-state resistance			5.5		mΩ
R _{DS(on)}	Drain-source on-state resistance	I _D =20A, V _{GS} =4.5V		7.0	10	mΩ
Dynamic Electrical Characteristics (unless otherwise stated) @ T_J=25°C						
C _{iss}	Input capacitance	f=1MHz, V _{DS} =30V V _{GS} =0V	1660	1950	2240	pF
C _{oss}	Output capacitance	f=1MHz, V _{DS} =30V V _{GS} =0V	715	840	965	pF
C _{rss}	Reverse transfer capacitance	f=1MHz, V _{DS} =30V V _{GS} =0V	25	35	45	pF
R _g	Gate resistance	f=1MHz		1.3		Ω
Q _g (10V)	Total gate charge	I _D =40A, V _{DS} =30V V _{GS} =10V		30		nC
Q _g (4.5V)	Total gate charge	I _D =40A, V _{DS} =30V V _{GS} =10V		14		nC
Q _{gs}	Gate-source charge	I _D =40A, V _{DS} =30V V _{GS} =10V		6.8		nC
Q _{gd}	Gate-drain charge	I _D =40A, V _{DS} =30V V _{GS} =10V		4.1		nC
Switching Characteristics						
t _{D(on)}	Turn-on delay time	I _D =40A, R _G =3Ω V _{DD} =30V, V _{GS} =10V		8.8		ns
t _r	Turn-on rise time	I _D =40A, R _G =3Ω V _{DD} =30V, V _{GS} =10V		55		ns
t _{D(off)}	Turn-off delay time	I _D =40A, R _G =3Ω V _{DD} =30V, V _{GS} =10V		23		ns
t _f	Turn-off fall time	I _D =40A, R _G =3Ω V _{DD} =30V, V _{GS} =10V		33		ns
Source- Drain Diode Characteristics (unless otherwise stated) @ T_J=25°C						
V _{SD}	Forward on voltage	I _{SD} =40A, V _{GS} =0V		0.8	1.2	V
t _{rr}	Reverse recovery time(T _J =25°C)	di/dt=100A/μs, I _{SD} =40A V _{GS} =0V		27		ns
Q _{rr}	Reverse recovery charge(T _J =25°C)	di/dt=100A/μs, I _{SD} =40A V _{GS} =0V		14		nC

NOTE:

- ① Repetitive rating; pulse width limited by max junction temperature.
- ② The power dissipation PDSM is based on R θ JA and the maximum allowed junction temperature of 150°C.
- ③ Limited by TJmax, starting TJ = 25°C, L = 0.5mH, RG = 25 Ω , IAS = 10A, VGS = 10V. Part not recommended for use above this value



Switching Test Circuit and Switching Waveforms

Characteristics Curve

Figure.1 Typical output characteristics

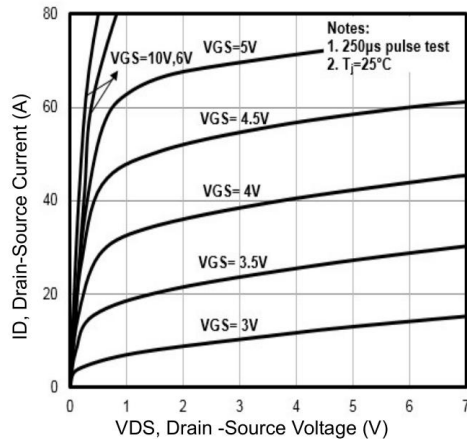


Figure.2 Typical transfer characteristics

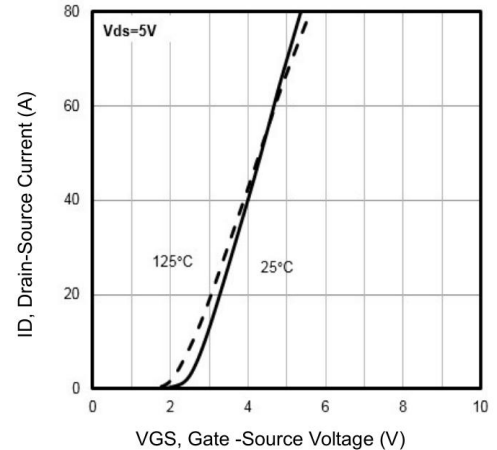


Figure.3 Typical normalized on-resistance vs. T_J

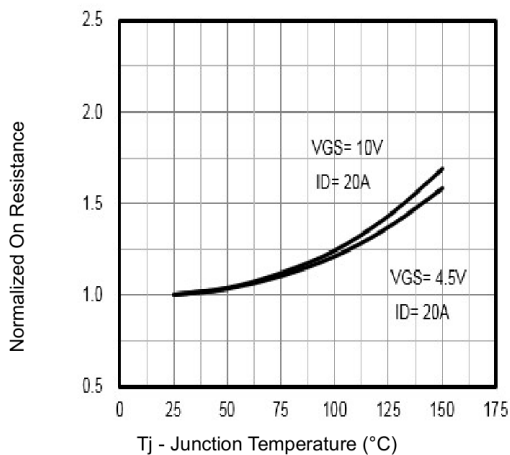


Figure.4 Typical source-drain diode forward voltage

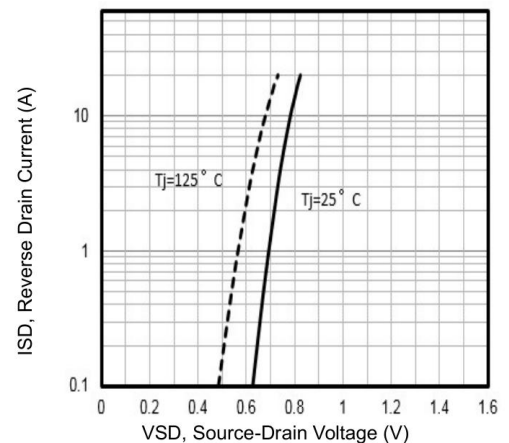


Figure.5 Maximum safe operating area

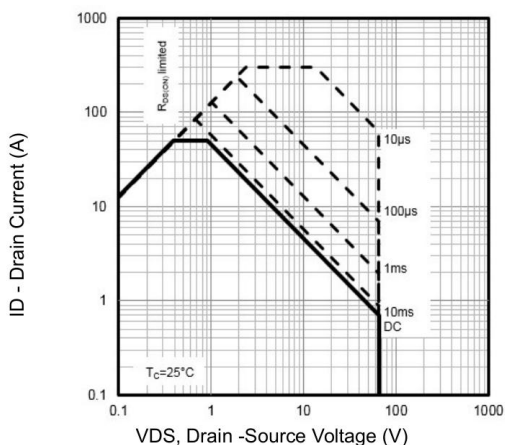


Figure.6 Typical capacitance vs. Drain-source voltage

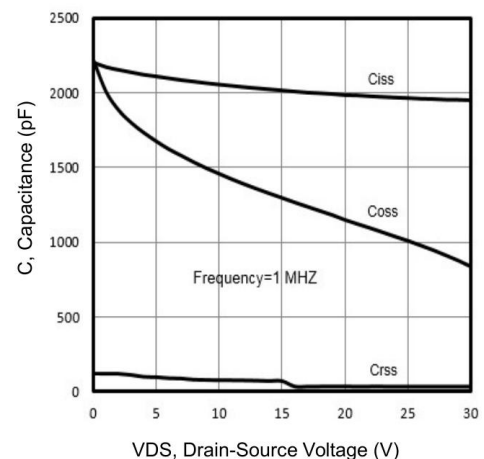


Figure.7 Typical gate charge vs. Gate-source voltage

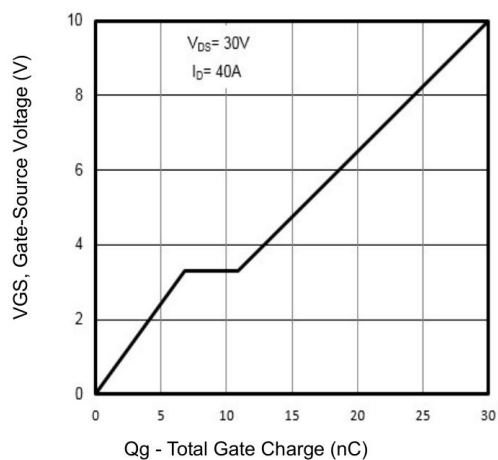


Figure.8 VGS(th) gate-source voltage vs. T_j

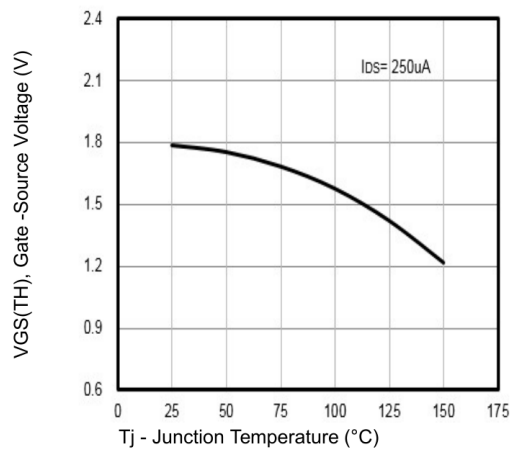
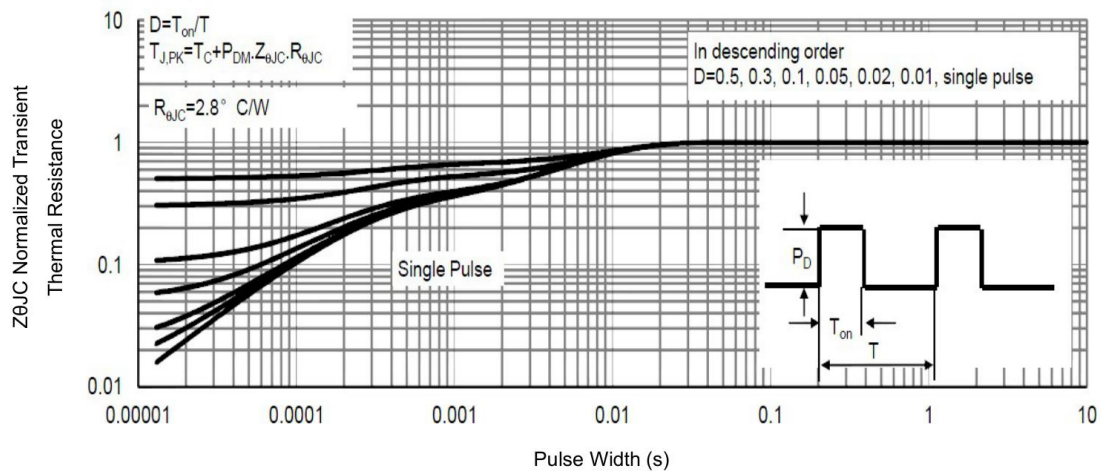
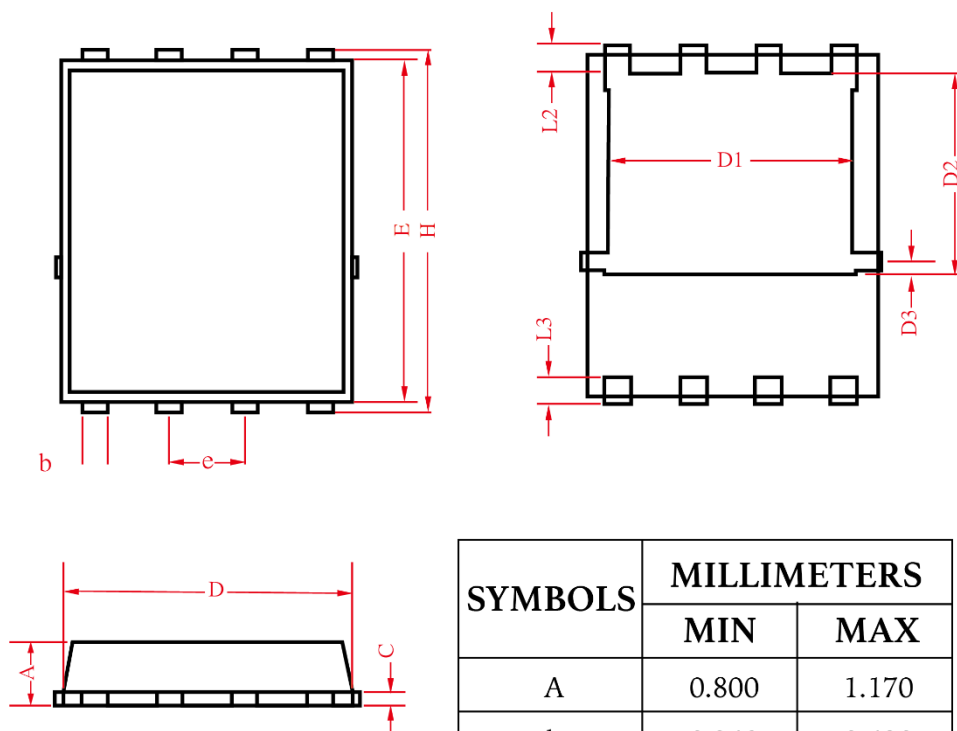


Figure.9 Normalized maximum transient thermal impedance



PDFN5*6 PACKAGE OUTLINE DIMENSIONS



SYMBOLS	MILLIMETERS	
	MIN	MAX
A	0.800	1.170
b	0.340	0.490
c	0.20	0.34
D	4.800	5.100
D1	3.800	4.200
D2	3.180	3.78
D3	0.150	0.360
E	5.650	5.900
e	1.270 TYP	
H	5.900	6.150
L1	0.050	0.250
L2	0.380	0.620
L3	0.380	0.75



Notice

1. Specification of the products displayed herein is subject to change without notice. Continuous development may necessitate change in technical data without notice. GEM-MICRO or anyone on its behalf assumes no responsibility or liability for any errors or inaccuracies.
2. Stresses beyond those listed under “Absolute Maximum Ratings” may cause permanent damage to the device. These are stress ratings only and functional operation of the device at these or any other conditions beyond those indicated in the operational sections of the specifications are not implied. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.
3. Details are subject to change without notice .